

BALL GRID ARRAY (BGA)

Types, Assembly & Soldering Complete Guide

PBGA • CBGA • FBGA • TBGA • μ BGA • CSP • WLCSP
Solder Balls • Reflow • Underfill • Rework • X-Ray Inspection

PCBA Manufacturing & Advanced Packaging Standards
IPC-7093 | IPC/J-STD-001 | IPC-A-610 | JEDEC

1. BGA OVERVIEW

Ball Grid Array (BGA) is a surface-mount packaging technology where the component's I/O connections are arranged in a grid pattern on the bottom of the package, using solder balls instead of peripheral leads (as in QFP or SOIC). This design dramatically increases I/O density, improves electrical performance, and simplifies PCB routing.

Why BGA Replaced QFP:

- **Higher I/O density:** 1000+ pins in a compact footprint vs. 200–300 for QFP
- **Shorter signal paths:** Balls connect directly to internal traces, reducing inductance
- **Better electrical performance:** Lower parasitic inductance and capacitance
- **Easier PCB routing:** Grid allows escape routing on inner layers
- **Improved solder joint reliability:** Larger contact area vs. fine-pitch gull-wing leads
- **Self-alignment:** Surface tension pulls component into alignment during reflow

BGA Challenges:

- **Hidden joints:** Solder balls are not visible for optical inspection
- **Expensive inspection:** X-ray required for quality verification
- **Rework difficulty:** Requires specialized hot-air stations and skilled technicians
- **Thermal management:** High-power BGAs need heat sinks or thermal vias
- **CTE mismatch:** Different expansion rates between package and PCB cause stress

■ **In-House Note:** BGA assembly is standard for in-house PCBA operations. All common types (PBGA, FBGA, CBGA, TBGA, μ BGA) are handled with standard reflow equipment. WLCSP and micro-BGA require underfill (LOCTITE 3549) for automotive reliability.

2. BGA PACKAGE TYPES

2.1 PBGA — Plastic Ball Grid Array

PBGA is the most common BGA type, using a **laminated organic substrate** (BT resin or FR-4) with solder balls attached to the bottom. The die is wire-bonded to the substrate and encapsulated in plastic molding compound.

Key Characteristics:

- **Substrate:** Organic laminate (BT resin, FR-4)
- **Ball material:** SAC305 (lead-free) or Sn63Pb37 (leaded legacy)
- **Ball pitch:** 1.0 mm, 1.27 mm (standard); 0.8 mm (fine-pitch)
- **Body size:** 15–45 mm typical
- **I/O count:** 144–1156 balls typical
- **Die attach:** Wire-bond (gold or copper)
- **Encapsulation:** Epoxy molding compound (EMC)
- **Cost:** Low to moderate — most economical BGA type

Applications:

- Microprocessors and chipsets (Intel, AMD)
- FPGA and DSP devices
- Memory controllers and networking ICs
- Consumer electronics (game consoles, set-top boxes)

2.2 CBGA — Ceramic Ball Grid Array

CBGA uses a **ceramic substrate** (alumina Al_2O_3 or aluminum nitride AlN) instead of organic laminate. Ceramic provides superior thermal conductivity and hermetic sealing for high-reliability applications.

Key Characteristics:

- **Substrate:** Ceramic (Al_2O_3 or AlN)
- **Ball material:** 90Pb10Sn (high-lead) or SAC305
- **Ball pitch:** 1.0 mm, 1.27 mm
- **Body size:** 25–50 mm typical
- **I/O count:** 256–1657 balls
- **Die attach:** Flip-chip or wire-bond
- **CTE:** ~6–7 ppm/°C (closer to silicon than organic)
- **Cost:** High — ceramic material and processing

Applications:

- High-performance microprocessors (IBM PowerPC, Sun SPARC)
- Aerospace and military electronics

- High-frequency RF modules
- Power devices requiring superior thermal dissipation

2.3 FBGA — Fine-pitch Ball Grid Array

FBGA (also called μ **BGA** or **Micro-BGA**) is a smaller BGA variant with **fine ball pitch** (≤ 0.8 mm) used for memory devices, ASICs, and compact logic devices.

Key Characteristics:

- **Substrate:** Thin organic laminate
- **Ball material:** SAC305 or SnAgCu variants
- **Ball pitch:** 0.4–0.8 mm
- **Body size:** 4–15 mm
- **I/O count:** 48–256 balls
- **Die attach:** Wire-bond or flip-chip
- **Height:** 0.8–1.4 mm (thin profile)
- **Cost:** Moderate

Applications:

- DDR SDRAM and Flash memory
- Mobile application processors
- Camera modules and image sensors
- Small form-factor logic devices

2.4 TBGA — Tape Ball Grid Array

TBGA uses a **polyimide tape substrate** with a copper heat spreader. The flexible tape allows very thin profiles and excellent thermal performance for mobile and compact devices.

Key Characteristics:

- **Substrate:** Polyimide tape (flexible)
- **Heat spreader:** Copper (typically coated with black epoxy)
- **Ball material:** SAC305 or SnPb
- **Ball pitch:** 1.0–1.27 mm
- **Body size:** 20–40 mm
- **I/O count:** 200–600 balls
- **Die attach:** Wire-bond to tape
- **Height:** 1.0–1.5 mm (very thin)
- **Cost:** Moderate to high

Applications:

- Notebook processors (Intel Mobile Pentium)
- Graphics processors (NVIDIA, ATI mobile GPUs)
- High-density memory modules
- Compact networking devices

2.5 μ BGA — Micro Ball Grid Array

μ BGA is the smallest standard BGA format, with body sizes as small as 2–5 mm and ball pitches of 0.4–0.5 mm. It bridges the gap between standard BGA and CSP/WLCSP.

Key Characteristics:

- **Substrate:** Thin organic or ceramic
- **Ball material:** SAC305 (lead-free mandatory)
- **Ball pitch:** 0.4–0.5 mm
- **Body size:** 2–8 mm
- **I/O count:** 16–64 balls
- **Die attach:** Flip-chip preferred
- **Height:** 0.5–1.0 mm
- **Cost:** Moderate

Applications:

- Camera modules (CMOS image sensors)
- RF transceivers and power management ICs
- Bluetooth and Wi-Fi modules

- Wearable device processors

2.6 Comparison Table

Type	Substrate	Ball Pitch	Body Size	I/O	Cost	Best For
PBGA	Organic (BT/FR-4)	1.0–1.27 mm	15–45 mm	144–1156	Low	General purpose, consumer
CBGA	Ceramic (Al ₂ O ₃ /AlN)	1.0–1.27 mm	25–50 mm	256–1657	High	High-reliability, military
FBGA	Thin organic	0.4–0.8 mm	4–15 mm	48–256	Moderate	Memory, mobile ASICs
TBGA	Polyimide tape	1.0–1.27 mm	20–40 mm	200–600	Moderate-High	Mobile processors, thin profile
μBGA	Thin organic/ceramic	0.4–0.5 mm	2–8 mm	16–64	Moderate	Camera, RF, wearables
CSP	Organic/ceramic	0.5–1.0 mm	3–15 mm	16–300	Moderate	Compact logic, sensors
WLCSP	Silicon die only	0.35–0.5 mm	1–3 mm	4–100	Low-Moderate	Mobile PMIC, ultra-compact

Table 1: BGA Package Types Comparison

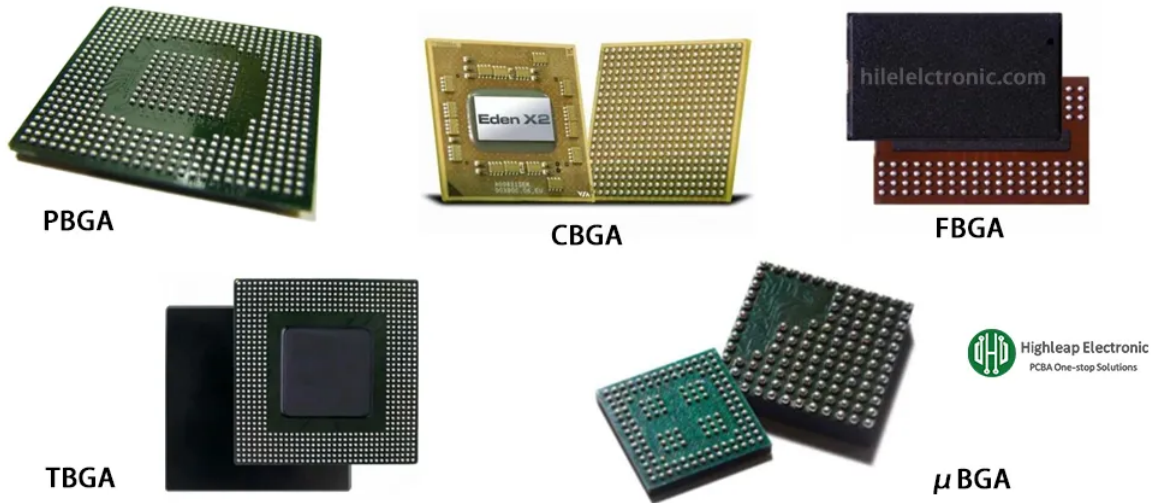


Figure 1: BGA Package Types — PBGA, CBGA, FBGA, TBGA, μBGA

3. BGA SOLDER BALLS

3.1 Ball Materials

BGA solder balls are the critical mechanical and electrical interface between the package and PCB. The material choice affects reliability, thermal performance, and RoHS compliance.

SAC305 (Sn96.5/Ag3.0/Cu0.5):

- **Status:** Industry standard lead-free alloy
- **Melting point:** 217–220°C
- **Reliability:** Good thermal cycling; moderate drop-shock
- **RoHS:** Fully compliant
- **Use:** PBGA, FBGA, μBGA, CSP

SAC387 (Sn95.5/Ag3.8/Cu0.7):

- **Status:** Early RoHS alloy; higher silver than SAC305
- **Melting point:** 217°C
- **Reliability:** Better creep resistance than SAC305
- **Cost:** Higher due to extra silver

Sn63Pb37 (Tin-Lead):

- **Status:** Legacy alloy; RoHS restricted
- **Melting point:** 183°C
- **Reliability:** Excellent long-term data; superior drop-shock
- **Use:** Exempt applications only (medical, military, aerospace)

90Pb10Sn (High-Lead):

- **Status:** High-temperature alloy for CBGA
- **Melting point:** 268–302°C
- **Reliability:** Excellent for high-temp applications
- **Use:** CBGA, high-power devices, step-soldering

SnAg (Tin-Silver, no Copper):

- **Status:** Alternative lead-free; avoids copper dissolution
- **Melting point:** 221°C
- **Use:** Applications sensitive to copper pad erosion

3.2 Ball Diameters by Pitch

Ball Pitch	Ball Diameter	Typical Use	Pad Diameter (PCB)

1.27 mm	0.75–0.89 mm	Standard PBGA, CBGA	0.60–0.70 mm
1.0 mm	0.60–0.76 mm	Fine-pitch BGA	0.50–0.60 mm
0.8 mm	0.46–0.56 mm	FBGA, fine-pitch	0.40–0.45 mm
0.65 mm	0.40–0.46 mm	Compact CSP	0.30–0.35 mm
0.5 mm	0.30–0.35 mm	μBGA, WLCSP	0.25–0.30 mm
0.4 mm	0.25–0.30 mm	Ultra-fine μBGA	0.20–0.25 mm

Table 2: BGA Ball Diameter vs. Pitch

4. BGA ASSEMBLY PROCESS

4.1 Solder Paste Printing

BGA solder paste printing is critical — the paste volume determines joint reliability, voiding, and coplanarity.

Stencil Design:

- **Aperture:** Round, 1:1 to pad size (no oversize for BGA)
- **Stencil thickness:** 100–150 μm (standard); 80–100 μm for 0.4–0.5 mm pitch
- **Area ratio:** ≥ 0.66 (aperture area / wall area)
- **Step-down stencil:** Use 50–70 μm step for 0.4 mm pitch μBGA
- **Electroformed stencil:** Recommended for fine-pitch (< 0.5 mm)

Paste Requirements:

- **Alloy:** Match BGA ball alloy (SAC305 for lead-free)
- **Powder type:** Type IV (20–38 μm) for 0.5–0.8 mm pitch
- **Powder type:** Type V (15–25 μm) for 0.4 mm pitch
- **Flux:** No-clean ORH1 for standard; ORH0 for sensitive applications

4.2 Component Placement

BGA placement accuracy is less critical than QFP due to self-alignment, but gross misalignment causes bridging or open joints.

- **Placement accuracy:** ± 50 μm at 3σ (standard BGA)
- **Placement accuracy:** ± 25 μm at 3σ (μBGA , WLCSP)
- **Placement force:** 1.5–3.0 N (too high damages balls)
- **Vision recognition:** Ball pattern or package edge recognition
- **Nozzle:** 3–5 mm rubber or ceramic nozzle (size matched to package)

4.3 Reflow Soldering

BGA reflow requires controlled profiles to ensure all balls melt simultaneously and self-alignment occurs properly.

Standard SAC305 BGA Reflow Profile:

- **Preheat:** 25 $\rightarrow$ 150°C at 1.5–2.5°C/sec
- **Soak:** 150 $\rightarrow$ 180°C for 60–90 seconds
- **Ramp to peak:** 180 $\rightarrow$ 245°C at 1.5–2.0°C/sec
- **Peak:** 245–250°C
- **TAL:** 60–90 seconds above 217°C
- **Cooling:** 2–4°C/sec down to 100°C

Key BGA Reflow Rules:

- **Nitrogen atmosphere:** Mandatory for 0.5 mm pitch and below
- **Board support:** Use pins or fixture to prevent warpage
- **Conveyor speed:** Matched to thermal mass (slower for large BGAs)
- **Zone count:** Minimum 8 zones for large BGA; 10+ zones preferred

4.4 Self-Alignment Mechanism

BGA packages **self-align** during reflow due to surface tension of molten solder. As long as the placement offset is < 50% of ball pitch, the surface tension pulls the package into proper alignment. This is a major advantage over QFP, which requires precise placement.

- **Self-alignment limit:** ~50% of ball pitch
- **Example:** 1.0 mm pitch BGA tolerates ± 0.5 mm placement offset
- **Example:** 0.5 mm pitch μ BGA tolerates ± 0.25 mm offset
- **Caution:** Excessive paste or contamination prevents self-alignment

5. BGA UNDERFILL

5.1 Purpose of Underfill

Underfill is a liquid epoxy dispensed beneath the BGA after reflow. It flows by capillary action and cures to form a rigid mechanical bond between the package and PCB. Underfill is mandatory for high-reliability applications.

Benefits:

- **Mechanical reinforcement:** Distributes stress across entire package area
- **Thermal cycling improvement:** 10–100× improvement in solder joint fatigue life
- **Drop-shock protection:** Prevents ball cracking under impact
- **Moisture barrier:** Reduces moisture ingress to solder joints
- **CTE mismatch compensation:** Absorbs differential expansion between package and PCB

5.2 In-House Underfill: LOCTITE 3549

LOCTITE® 3549 is the in-house underfill material documented in the Adhesive Guide. It is a one-component, heat-cure epoxy designed for CSP and BGA underfill.

- **Type:** Capillary flow underfill
- **Cure:** 150°C for 10–15 minutes
- **CTE:** Matched to solder and PCB materials
- **Storage:** Frozen -40°C; 30-day shelf life at 25°C after thaw

5.3 Underfill Process

1. **Complete reflow:** Solder BGA first, verify X-ray
2. **Clean (optional):** Remove flux residue if water-soluble flux used
3. **Preheat board:** 40–60°C (reduces viscosity, speeds flow)
4. **Dispense:** L-shape or I-shape along package edge
5. **Flow:** 30–120 seconds for complete underfill
6. **Inspect fillet:** Verify complete edge coverage
7. **Cure:** 150°C convection oven for 10–15 minutes
8. **Verify:** C-SAM (Acoustic Microscopy) for void detection (Class 3)

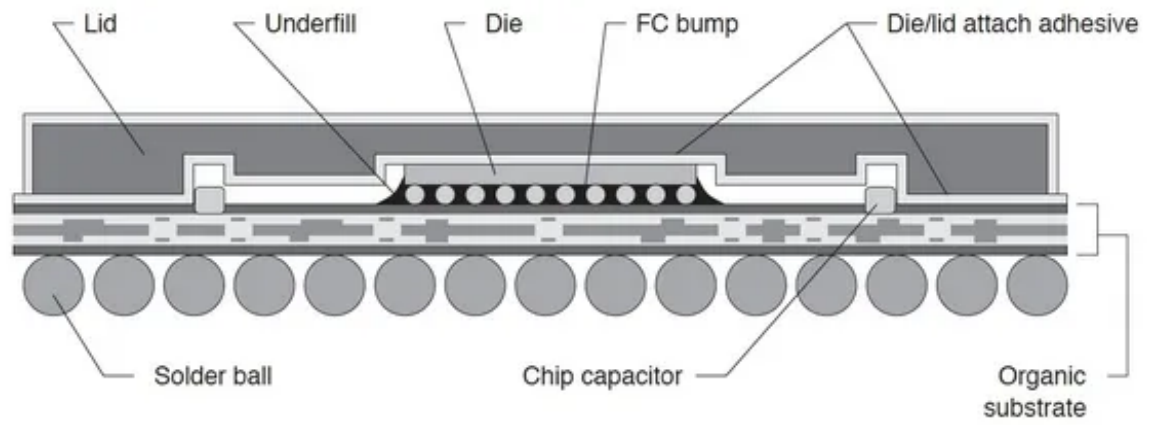


Figure 2: BGA Cross-Section — Lid, Underfill, Die, FC Bump, Solder Ball, Organic Substrate

6. BGA INSPECTION METHODS

6.1 X-Ray Inspection

X-ray is the primary inspection method for BGA because solder joints are hidden under the package. Both 2D and 3D X-ray systems are used.

2D X-Ray (Transmission):

- **Principle:** Single-angle X-ray image showing ball projection
- **Detects:** Bridging, missing balls, gross voids, ball displacement
- **Limitation:** Cannot detect head-in-pillow or hidden voids in multi-layer boards
- **Cost:** Lower than 3D X-ray

3D X-Ray (AXI — Automated X-ray Inspection):

- **Principle:** Multiple angles reconstructed into 3D model
- **Detects:** Head-in-pillow, void volume, crack propagation, ball shape
- **Advantage:** Complete volumetric analysis of each joint
- **Cost:** High — premium inspection equipment

6.2 Voiding Analysis

Voids (air pockets) in BGA solder joints are common and generally acceptable within limits. Excessive voiding reduces thermal and electrical conductivity.

- **IPC-A-610 Class 2:** $\leq 25\%$ void area per ball
- **IPC-A-610 Class 3:** $\leq 15\%$ void area per ball
- **Typical voiding:** 5–15% for well-controlled processes
- **Cause of high voiding:** Insufficient flux, excessive peak temp, poor pad design
- **Mitigation:** Optimize flux chemistry, use N₂ atmosphere, design thermal vias

6.3 Acoustic Microscopy (C-SAM)

C-Mode Scanning Acoustic Microscopy (C-SAM) uses ultrasound to detect delamination, voids, and cracks in packages and underfill. It is the gold standard for underfill verification.

- **Detects:** Underfill voids, delamination, die attach defects
- **Resolution:** $\sim 10\text{--}50\ \mu\text{m}$
- **Use case:** 100% inspection for automotive/military BGAs with underfill
- **Limitation:** Requires water coupling; boards must be dry

6.4 Electrical Testing

- **Boundary Scan (JTAG):** Tests interconnections without physical probe access
- **ICT (In-Circuit Test):** Bed-of-nails testing for manufacturing defects
- **Functional Test:** Validates device operation under load

7. BGA REWORK PROCESS

7.1 Rework Equipment

BGA rework requires specialized hot-air stations with precise temperature control and component handling.

- **Hot-air rework station:** Bottom + top heaters with profile control
- **Nozzle:** Custom size matched to BGA body (3–50 mm range)
- **Vacuum pickup:** For component removal and replacement
- **Preheater:** Bottom-side board heating to reduce thermal shock
- **Alignment system:** Split-prism or camera-based alignment

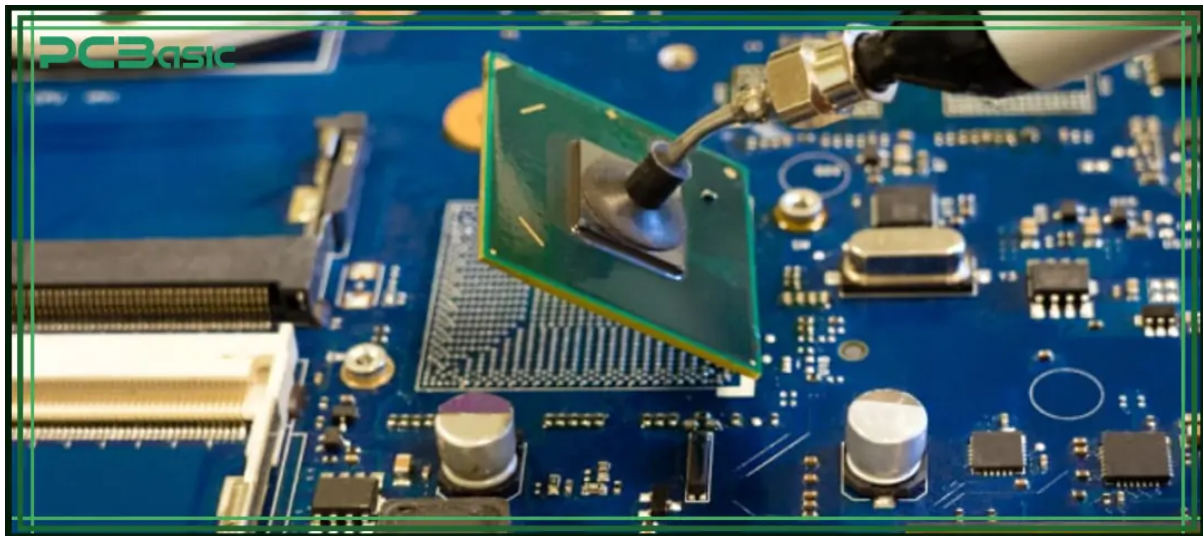


Figure 3: BGA Rework Process — Hot Air Removal and Replacement

7.2 Rework Procedure

1. **Preparation:** Secure board in fixture; apply Kapton tape to protect nearby components
2. **Preheat:** Heat board from bottom to 100–150°C
3. **Remove old BGA:** Apply hot air at 250–280°C; lift with vacuum when solder melts
4. **Clean pads:** Remove old solder with solder wick and flux; verify flatness
5. **Apply paste:** Dispense solder paste or place solder preforms on pads
6. **Place new BGA:** Align using split-prism or camera system
7. **Reflow:** Apply profile matching original assembly; allow self-alignment
8. **Cool:** Controlled cooling to prevent thermal shock
9. **Inspect:** X-ray verification; electrical test

7.3 Reballing

Reballing is the process of replacing solder balls on a used or defective BGA package. It is common for component recovery and repair operations.

Reballing Methods:

- **Stencil method:** Place BGA on reballing stencil; apply solder paste; reflow

- **Preform method:** Place solder sphere preforms on each pad; reflow
- **Ball attach machine:** Automated placement of individual solder balls

Reballing Considerations:

- **Bake before reball:** 125°C for 24 hours to remove moisture
- **Ball alloy:** Match original or customer specification
- **Reflow profile:** Lower peak than PCB assembly to protect die
- **Inspection:** Verify ball coplanarity (< 0.1 mm)



Figure 4: BGA Reballing Equipment — Alignment System and Solder Ball Placement

8. COMMON BGA DEFECTS

8.1 Head-in-Pillow (HiP)

The BGA solder ball partially collapses but does not fully bond to the PCB pad. The ball looks like a 'head resting on a pillow' in X-ray.

Causes:

- PCB or BGA warpage during reflow
- Oxidation on ball or pad surface
- Insufficient flux activity
- Contamination on pad (oils, fingerprints)

Solutions:

- Reduce peak temperature; extend TAL
- Use board support pins to minimize warpage
- Use higher activity flux (ORH1)
- Ensure clean, dry pads before paste print
- Nitrogen atmosphere mandatory for fine-pitch

8.2 Bridging

Solder bridges between adjacent balls, causing short circuits.

Causes:

- Excessive solder paste volume
- Misaligned stencil print
- Component placement offset
- Collapsed balls (excessive reflow temperature)

Solutions:

- Reduce stencil aperture by 10%
- Verify print accuracy $\pm 25 \mu\text{m}$
- Reduce peak temperature to prevent ball collapse

8.3 Open Joints

No solder connection between ball and pad.

Causes:

- Insufficient solder paste
- Pad contamination (non-wetting)
- BGA lifted during reflow (tombstoning on corner balls)

- Missing ball from package

Solutions:

- Increase stencil aperture or thickness
- Verify pad plating quality (ENIG, OSP)
- Check BGA ball coplanarity before placement

8.4 Voiding

Air pockets trapped in solder joint.

Causes:

- Flux outgassing during reflow
- Insufficient flux to wet large thermal pads
- Poor pad design (no thermal vias)

Solutions:

- Optimize flux chemistry for low voiding
- Use nitrogen atmosphere
- Design thermal vias in center pad (0.3 mm pitch)

8.5 Cracked Solder Joints

Mechanical or thermal stress fractures in solder.

Causes:

- Excessive board flexure during handling or test
- CTE mismatch between package and PCB
- Thermal cycling without underfill

Solutions:

- Apply underfill (LOCTITE 3549) for high-stress applications
- Use compliant solder alloys (SAC105 for drop-shock)
- Support board during ICT and handling

■ **In-House Quality Target:** For automotive BGA assemblies (Tesla, Zoox), target < 5% voiding, zero bridging, zero open joints, and 100% X-ray inspection. Implement SPC tracking of void percentage by BGA lot number.

9. QUICK REFERENCE SUMMARY

BGA Types:

- **PBGA:** Organic substrate, low cost, general purpose
- **CBGA:** Ceramic substrate, high reliability, military/aerospace
- **FBGA:** Fine pitch, thin profile, memory and mobile ASICs
- **TBGA:** Tape substrate, thin profile, mobile processors
- **μBGA:** Micro size, 0.4–0.5 mm pitch, camera/RF modules
- **CSP:** $\leq 1.2\times$ die size, compact logic
- **WLCSP:** Die-size package, 0.35–0.5 mm pitch, mobile PMIC

Solder Balls:

- **SAC305:** Standard lead-free, 217°C, RoHS compliant
- **Sn63Pb37:** Legacy, 183°C, exempt use only
- **90Pb10Sn:** High-lead, 268°C, CBGA/high-temp
- **Ball diameter:** 0.25–0.89 mm depending on pitch

Assembly Process:

- **Stencil:** Round aperture, 1:1 to pad, 100–150 μm thick
- **Paste:** Type IV/V, match ball alloy, no-clean ORH1
- **Placement:** $\pm 50\text{ }\mu\text{m}$ (standard), $\pm 25\text{ }\mu\text{m}$ (fine-pitch)
- **Reflow:** Peak 245–250°C, TAL 60–90 sec, N $\blacksquare$ for $< 0.5\text{ mm}$ pitch
- **Self-alignment:** Works up to 50% of ball pitch offset

Underfill:

- **LOCTITE 3549** — Capillary flow, 150°C cure, frozen storage
- **Mandatory for:** Automotive, mobile, thermal cycling applications
- **Process:** Dispense edge → flow 30–120 sec → cure 150°C/10–15 min
- **Verify:** C-SAM for voids (Class 3)

Inspection:

- **X-ray 2D:** Bridging, missing balls, gross voids
- **X-ray 3D (AXI):** Head-in-pillow, void volume, cracks
- **C-SAM:** Underfill voids, delamination
- **Void limits:** $\leq 25\%$ (Class 2); $\leq 15\%$ (Class 3)

Rework:

- **Equipment:** Hot-air station with bottom + top heaters
- **Removal:** Preheat 100–150°C; hot air 250–280°C; vacuum lift

- **Reballing:** Stencil or preform method; bake before reball
- **Verify:** X-ray and electrical test after rework

Common Defects:

- **Head-in-Pillow:** Reduce peak temp, board support, N■
- **Bridging:** Reduce paste, verify print accuracy
- **Open joints:** Increase paste, verify pad plating
- **Voiding:** Optimize flux, N■, thermal vias
- **Cracks:** Underfill, compliant alloy, board support

References & Standards:

- IPC-7093: Design and Assembly Process Implementation for Bottom Termination Components
- IPC/J-STD-001: Requirements for Soldered Electrical and Electronic Assemblies
- IPC-A-610: Acceptability of Electronic Assemblies
- JEDEC J-STD-012: Implementation of Flip Chip and Chip Scale Technology
- JEDEC JESD22-B117: BGA Solder Ball Shear Test Method
- IPC-TM-650 2.4.43: BGA Solder Ball Pull Test
- IPC-TM-650 2.4.35: BGA Pad Cratering Test
- MIL-STD-883: Test Methods and Procedures for Microelectronics
- Henkel LOCTITE Technical Datasheet: 3549 Underfill